

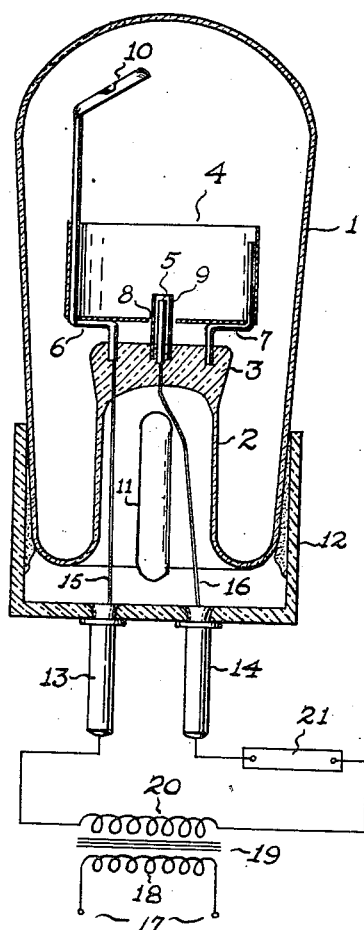
June 1, 1937.

J. D. LE VAN

2,082,165

GASEOUS DISCHARGE RECTIFIER

Original Filed Oct. 8, 1932



INVENTOR

JAMES D. LE VAN

by

Elmer J. Gorn
ATTORNEY

UNITED STATES PATENT OFFICE

2,082,165

GASEOUS DISCHARGE RECTIFIER

James D. Le Van, Belmont, Mass., assignor to
Raytheon Production Corporation, Newton,
Mass., a corporation of Delaware

Application October 8, 1932, Serial No. 636,852
Renewed October 30, 1936

8 Claims. (Cl. 250—27.5)

This invention relates to a gaseous discharge rectifier, and more particularly to a rectifier in which a glow discharge takes place between a cathode having an extended area and an anode of limited area.

An object of my invention is to provide such a rectifier having a low voltage drop, long life, and low value of back current.

The foregoing and other objects of my invention will be best understood from the following description of an exemplification thereof, reference being had to the accompanying drawing, wherein:

The single figure is a cross-sectional view of an embodiment of my invention, together with a diagrammatic representation of a circuit in which said invention may be used.

The device as illustrated consists of an envelope 1, which may be constructed of glass or the like, containing a gas preferably of the monatomic group, for example helium. The envelope has a reentrant stem 2 carrying a press 3 at its upper end. The press 3 supports a cathode 4 and an anode 5. The cathode 4 is preferably constructed of thin copper drawn into the shape of a cup. Of course any other shape which gives an extended operating surface for the anode could likewise be used. The cathode 4 is supported from the press 3 by means of two wire standards 6 and 7 sealed at their lower ends into said press 3 and securely fastened to the sides of the cathode 4 by some suitable means, such as, for example, being welded thereto. The anode 5 consists of a small metal rod of nickel, tantalum, iron, or any other suitable refractory metal, and is sealed at its lower end in the press 3. The cathode has formed in its lower wall an aperture or opening 8 through which the anode 5 may actually project or be merely presented to the interior of the cathode. In order to limit the discharge to the end of the anode and the interior of the cathode, I provide a sleeve or shield 9 sealed in the press, surrounding the anode 5 and projecting through the opening 8 in the cathode. This sleeve is preferably formed of a metal, such as specified for the anode. There is no metallic conductive connection between this sleeve and either the anode or cathode. The diameters of the opening 8 in the cathode, the sleeve 9, and the anode 5 are such that the clearance between juxtaposed surfaces is preferably of the order of or less than the mean free path of electrons present in the gas, thus providing an insulating short space between each of these members so that a discharge can occur only as stated above.

In order to reduce the losses in my tube and to lower the voltage drop at the cathode, I provide my cathode with a novel electron-emissive coating. This coating is of the general type which includes the use of an alkaline earth metal. In order to form my novel coating, the tube may be arranged and treated as follows. One of the standards, for example 6, is extended above the cathode 4, and carries at its upper end a capsule 10 containing a mixture which liberates an alkali metal upon heating. The mixture is preferably calcium and caesium chloride. The caesium chloride could be replaced by rubidium chloride, or a mixture of both chlorides could also be used. The cathode is first heated by some suitable means, such as high frequency induction, before any evacuation of the envelope 1 is had. This heating continues until the cup begins to oxidize, which oxidation takes place roughly at about 600° C. The exhaust tube 11 is then connected to a suitable vacuum pump which is set in operation to evacuate the envelope 1. At the same time the tube is baked in an oven at 400° C. for about ten minutes. The cathode 4 is again heated by high frequency induction to about 750° C., at which all surplus gases are driven off said cathode. After this operation has taken place, a thin layer of cuprous oxide will remain on the copper of the cathode. The tube is then allowed to cool for about three minutes, and the capsule 10 is then heated by high frequency induction until the mixture in the capsule is "flushed". This flashing liberates the alkali metal within the capsule 10. The cuprous oxide having a very strong affinity for said alkali metal will react therewith, and cause a thin film of said metal to deposit on the cuprous oxide. After the alkali metal coating has been formed on the cathode, the tube is filled with an ionizable atmosphere, preferably a monatomic gas, such as helium, at a pressure sufficient to support an ionizing discharge between the cathode and anode. Such a pressure is preferably 20 mm. of mercury in the case of helium, although pressures lying between 10 mm. and 25 mm. of mercury are satisfactory. The exhaust tube 11 may then be sealed off. I have found that a cathode prepared in this manner has an exceptionally low voltage drop or voltage loss, together with an exceptionally long life. Such a coating is very stable even in the presence of comparatively heavy discharges which may occur in the device. In a typical tube having a cathode prepared as herein described, I have found that with a given value of load current, the inside area of the cathode could be figured approximately as

one square inch for each fifty milliamperes of peak load current.

I have found that silver in the place of copper gives equally good results. Thus whenever I mention copper, it is to be understood that silver may also be used. Instead of forming an oxide layer on the cathode, a layer of sulphide can also be used. The sulphide could be formed by any suitable method. I prefer, however, to form this sulphide by holding the cathode over a quantity of ammonium sulphide. There is enough hydrogen sulphide in the vapor arising from the ammonium sulphide to react with the copper, giving a thin coating of copper sulphide. After the coating has thus been formed and the cathode mounted within the envelope, the rest of the procedure is approximately the same as for the oxide coating as given above.

In many tubes of the general type involved, the voltage drop at the cathode has risen gradually during the life of the tube, thus greatly lowering the efficiency thereof. This has been due, at least partly, to the fact that some of the anode metal is vaporized during operation, and deposits upon the electron-emissive surfaces of the cathode, increasing its voltage drop and decreasing the life of the tube. In order to eliminate this effect, I mount my anode so that its upper end lies below the top of the sleeve 9. Thus any metal which is vaporized therefrom is deposited on the inner walls of the sleeve and cannot escape to be deposited upon the walls of the cathode. Since my device is to operate as a rectifier, it is desirable that the back current be kept as low as possible. I have found that by placing the anode wholly within the sleeve 9, as stated above, the back current is greatly reduced. Such an arrangement operates very satisfactorily, it being merely necessary to take care that the anode should be clean before it is inserted into the tube. I have also found that the sleeve 9 formed of metals such as specified, does not vaporize to any appreciable degree, and therefore does not tend to increase the cathode drop.

The tube may be provided with the usual insulating base 12 carrying the contact prongs 13 and 14. A conductor 15 leads from one of the cathode standards 6 to one of the contact prongs 13, and a conductor 16 leads from the anode 5 to the other contact prong 14. The tube may be connected in any suitable circuit, preferably a rectifying circuit which may be, for example, such as that shown diagrammatically in the drawing. This circuit consists of a source of A. C. 17 connected to the primary 18 of a transformer 19. The secondary 20 of said transformer has one terminal thereof connected to one of the contact prongs 13. The other terminal of said secondary 20 is connected through a suitable output or load device 21 to the other of said contact prongs 14. Upon supplying A. C. from the source 17, a rectified direct current will flow through said output device 21.

This invention is not limited to the particular details of construction, materials or processes described above, as many equivalents will suggest themselves to those skilled in the art. It is accordingly desired that the appended claims be given a broad interpretation commensurate with the scope of the invention within the art.

What is claimed is:

1. A gaseous discharge device comprising a hermetically-sealed envelope containing an ionizable gas, a glow discharge cathode of extended

area and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of copper, a layer of copper oxide thereon and an alkali metal deposited on said layer, said operating surface being electron-emitting, said cathode being provided with an aperture, a reentrant stem carrying a press, a rod-shaped anode sealed in said press and presented to said electron-emitting surface through said aperture, a metal sleeve surrounding said anode from said press to a point sufficiently above the upper end of said anode to prevent material vaporized from said anode from escaping from said sleeve and extending through said aperture, said sleeve being insulated from said cathode and anode and spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and anode.

2. A gaseous discharge device comprising a hermetically-sealed envelope containing an ionizable gas, a glow discharge cathode of extended area and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of an oxidizable metal, a layer of oxide thereon, and an alkali metal deposited on said layer, said operating surface being electron-emitting, said cathode being provided with an aperture, a reentrant stem carrying a press, a rod-shaped anode sealed in said press and presented to said electron-emitting surface through said aperture, a metal sleeve surrounding said anode from said press to a point sufficiently above the upper end of said anode to prevent material vaporized from said anode from escaping from said sleeve and extending through said aperture, said sleeve being insulated from said cathode and anode and spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and anode.

3. A gaseous discharge device comprising a hermetically-sealed envelope containing an ionizable gas, a glow discharge cathode of extended area and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of copper, a layer of a compound of copper thereon, and an alkali metal deposited on said layer, said operating surface being electron-emitting, said cathode being provided with an aperture, a reentrant stem carrying a press, a rod-shaped anode sealed in said press and presented to said electron-emitting surface through said aperture, and a metal sleeve surrounding said anode from said press to a point sufficiently above the upper end of said anode to prevent material vaporized from said anode from escaping from said sleeve and extending through said aperture, said sleeve being insulated from said cathode and anode and spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and anode.

4. A gaseous discharge device comprising a hermetically-sealed envelope containing an ionizable gas, a glow discharge cathode of extended area, and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface

formed of copper, a layer of oxide thereon and caesium deposited on said layer, said operating surface being electron emitting, said cathode being provided with an aperture, a reentrant stem carrying a press, a rod-shaped anode sealed in said press and presented to said electron-emitting surface through said aperture, and a metal sleeve surrounding said anode from said press to a point sufficiently above the upper end of said anode to prevent material vaporized from said anode from escaping from said sleeve and extending through said aperture, said sleeve being insulated from said cathode and anode and spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and anode.

5. A gaseous discharge device comprising a hermetically-sealed envelope containing an inert gas at a pressure of the order of 20 mm. of mercury, a glow discharge cathode of extended area and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of an oxidizable metal, a layer of oxide thereon and an alkali metal deposited on said layer, said operating surface being electron emitting, said cathode being provided with an aperture, a reentrant stem carrying a press, a rod-shaped anode sealed in said press and presented to said electron-emitting surface through said aperture, and a metal sleeve surrounding said anode, said sleeve being insulated from said cathode and anode and spaced from said anode an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and anode.

6. A gaseous discharge device comprising an envelope containing an ionizable gas, a glow discharge cathode, and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of a metal, a layer of a compound of said metal thereon, and an alkali metal deposited on said layer, said operating surface being electron-emitting, said cathode being provided with an aperture, a stem, an anode sealed

in said stem and presented to said electron-emitting surface through said aperture, a metal sleeve surrounding said anode from said stem to a point adjacent the upper end of said anode, said sleeve being spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and said anode.

7. A gaseous discharge device comprising an envelope containing an ionizable gas, a glow discharge cathode, and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of a metal, a layer of an oxide thereon, and an alkali metal deposited on said layer, said operating surface being electron-emitting, said cathode being provided with an aperture, a stem, an anode sealed in said stem and presented to said electron-emitting surface through said aperture, a metal sleeve surrounding said anode from said stem to a point adjacent the upper end of said anode, said sleeve being spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and said anode.

8. A gaseous discharge device comprising an envelope containing an ionizable gas, a glow discharge cathode, and an anode, said electrodes being adapted to support an ionizing discharge through said gas, said cathode having its operating surface formed of copper, a layer of copper oxide thereon, and an alkali metal deposited on said layer, said operating surface being electron-emitting, said cathode being provided with an aperture, a stem, an anode sealed in said stem and presented to said electron-emitting surface through said aperture, a metal sleeve surrounding said anode from said stem to a point adjacent the upper end of said anode, said sleeve being spaced from said anode and the sides of said aperture an insulatingly short distance, said sleeve being open at its upper end to allow a discharge to pass between said electron-emitting surface and said anode.

JAMES D. LE VAN.